

Exp No: 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51

OBJECTIVES

- To study GPIO port control in the AT89C51 microcontroller.
- To understand bit-addressable Special Function Registers (SFRs) using Embedded C.
- To control individual port pins using bit addressing in the AT89C51.

PROGRAM

```
#include <reg51.h>

sbit LED1 = P1^0;
sbit LED2 = P1^1;

void delay(void)
{
    unsigned int i, j;
    for(i = 0; i < 200; i++)
        for(j = 0; j < 1000; j++);
}

void main(void)
{
    P1 = 0x00;
    while(1)
    {
        LED1 = 1;
        LED2 = 0;
        delay();
        LED1 = 0;
        LED2 = 1;
        delay();
    }
}
```

}

OUTPUT

The screenshot displays the Keil uVision IDE interface for a project named "Vibration_Monitoring\Keil\GPIO_Experiment". The main window shows a C source file "main.c" with the following code:

```
1 #include <reg51.h>
2
3 sbit LED1 = P1^0;
4 sbit LED2 = P1^1;
5
6 void delay(void)
7 {
8     unsigned int i, j;
9     for(i = 0; i < 200; i++)
10         for(j = 0; j < 1000; j++);
11 }
12
13
14 void main(void)
15 {
16     P1 = 0x00;
17     while(1)
18     {
19         LED1 = 1;
20         LED2 = 0;
21         delay();
22
23         LED1 = 0;
24         LED2 = 1;
25         delay();
26     }
27 }
28
```

A "Parallel Port 1" dialog box is open, showing the configuration for Port 1. The P1 pins are 0-7, and the P1 value is 0xFE. The P1 pins are 0-7, and the P1 value is 0xFE.

The Registers window shows the following values:

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x03
r5	0xa1
r6	0x00
r7	0x75
Sys	
a	0x00
b	0x00
sp	0x09
sp_max	0x09
dptr	0x0000
PC	0x0006
status	2879864
sec	1439.932
psw	0x00

The Command window shows the following command:

```
Running with Code Size Limit: 2K
Load "C:\KeilProjects\Vibration_Monitoring\Keil\Objects\GPIO_Experiment"
```

The Call Stack - Locals window shows the following information:

Name	Location/Value	Type
i	0x0075	uint
j	0x00A1	uint
MAIN	C:0x081D	

The status bar at the bottom shows the simulation time: 11: 2762.69207900 sec, L13 C1, CAP. NUM SCRL OVR. R/W.